DUALPATH-CORE: A MULTI-AGENT HYBRID TTRPG ENGINE

MUD

SENIOR THESIS

@FB607 | 29 MAY 2026 | 12:00-13:00

PROBLEM & THE LANGUAGE CHALLENGE

HALLUCINATION: AI DUNGEON MASTERS FREQUENTLY INVENT RULES, MISCALCULATE HP, AND ALLOW ILLEGAL SPATIAL MOVEMENTS (LIKE WALKING THROUGH SOLID WALLS).

THE CHALLENGE: PLAYERS USE UNSTRUCTURED, CREATIVE LANGUAGE ("I SLIDE UNDER THE TABLE"). RIGID PROGRAMMING CANNOT INTERPRET THIS, REQUIRING AN AI. WE PROPOSE A HYBRID SYSTEM TO HANDLE CREATIVE LANGUAGE VIA AI, WHILE ROUTING MATH TO A HARD-CODED PYTHON ENGINE.

SYSTEM OVERVIEW: TWO-PATH ARCHITECTURE

THE SYSTEM INTERCEPTS PLAYER INPUT AND CATEGORIZES IT INTO STANDARD ACTIONS OR CREATIVE REQUESTS.

Intent Router (Regex & LLM Semantic Match)

Path A: Python Rules	Path B: LLM Arbiter
Attacks & Movement. 100% accurate math, 0 token cost.	Improvised actions. Assigns DC & specific status effects.

TOON SERIALIZATION: GAME STATE IS COMPRESSED INTO TOKEN-ORIENTED OBJECT NOTATION, SAVING ~40% TOKEN COSTS.

EXPLORATION ENGINE

- POINT-CRAWL GRAPH: ROOMS ARE CONNECTED NODES.
- HYBRID ROUTER: USES REGEX FOR SIMPLE COMMANDS (LOOK), AND AI FOR COMPLEX LANGUAGE.
- SPATIAL GUARDRAILS: PYTHON VERIFIES ROOM CONNECTIONS BEFORE AI ACTS, **STOPPING SPATIAL HALLUCINATIONS**.
- INFINITE LORE PREVENTION: SEPARATES LOGS TO PREVENT REPEATING TEXT.

COMBAT & INITIATIVE

- DYNAMIC TURN ORDER: INITIATIVE QUEUE (D20 + PHYS) SORTS PLAYERS AND ENEMIES DYNAMICALLY.
- ZONE-BASED POSITION: GRID-LESS ZONES (NEAR, MID, FAR). RANGE DISADVANTAGES ARE CALCULATED ENTIRELY BY PYTHON.

PLAYTEST CONSOLE CAPTURE

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> Player: I go to the next room
🔴 [SYSTEM] Invalid MOVE blocked: 'the next room'
> Player: Go to main shaft
[MOVE] → Main Shaft (Pass 1: Regex match, 0 tokens)
> Player: I blow the toxic gas away with wind magic
[PUZZLE_ATTEMPT] Routed to LLM Arbiter. Success!
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